# Supplementary Information: Every Natural Strengthening Examined Is False

Amin Abaee
Independent Researcher, Tehran, Iran
amin_abaee@ut.ac.ir • ORCID 0000-0002-0019-1842

Submitted: 14 November 2026

## Contents

**1 The $h^*$-observations through $n = 10$** **2**
1.1 The table . . . 2
1.2 Strictness decay and non-correlation . . . 3
1.3 The residual question . . . 3

**2 Computational provenance** **4**
2.1 The ledger . . . 4
2.2 The runs behind this paper . . . 4
2.3 Certificates and budgets . . . 6
2.4 Reproduction . . . 6

**3 A self-contained verification core** **7**
3.1 The identity and the runner formulation . . . 7
3.2 The sum-only equality, explicitly . . . 8
3.3 Hand checks of the headline witnesses . . . 8

# 1 The $h^*$-observations through $n = 10$

This section records the Ehrhart observations on the LR-zonotopes, extended here to $n = 9$ and $n = 10$ in both core families. The extension has a narrow purpose: establish whether the real-rootedness and log-concavity observed at $n \le 8$ continue, and whether the strictness decay and the non-correlation with covering margins persist. Both answers are yes. The section closes with the statement we regard as the residual question of independent interest.

## 1.1 The table

Table 1 lists, for each family and $n$: the $h^*$-polynomial, real-rootedness (verified *exactly* by Sturm-sequence sign-change counts on the square-free part of the integer polynomial, with the 45-decimal-place numerical verification and the companion-matrix check agreeing), log-concavity and Newton's inequalities (exact rational arithmetic), the minimum ratio $h_k^2/(h_{k-1}h_{k+1})$ (the *strictness* $S$ of log-concavity), the Newton floor at the minimizing $k$, namely $\big((k{+}1)(d{-}k{+}1)\big)/\big(k(d{-}k)\big)$, and the excess ratio $S$/floor. Every row's Ehrhart data was verified three ways (Eulerian expansion, standard inversion, and an independent depth-first coset count at $t = 1, 2$), the volume identity $e_{n-1} = \sum v_j$ was asserted exactly, and the loneliness identities $\lambda = 1/(n{+}1)$ (harmonic) and $\lambda = 2/(2n{+}1)$ (rung-2) were re-asserted exactly. The $n \le 8$ rows reproduce the program's records digit for digit; the $n = 9, 10$ rows extend them.

Table 1: $h^*$-polynomials of the LR-zonotopes, $n = 3..10$, both core families. "RR" = real-rooted (exact Sturm verification); "LC/Nw" = log-concave and Newton (exact rational arithmetic); $S = \min_k h_k^2/(h_{k-1}h_{k+1})$; $nS = n \cdot S$; "floor" = the Newton bound $\big((k{+}1)(d{-}k{+}1)\big)/\big(k(d{-}k)\big)$ at the minimizing $k$ (any real-rooted family has $S \ge$ floor there); $S/f = S/\text{floor}$. The $n \le 8$ rows reproduce the program's records digit for digit; the $n = 9, 10$ rows extend them.

| family | $n$ | $h^*(z)$ | RR | LC/Nw | $S$ | $nS$ | floor | $S/f$ |
|---|---|---|---|---|---|---|---|---|
| harm. | 3 | $1+7z+4z^2$ | y | y | 12.250 | 36.8 | 4 | 3.06 |
| rung2 | 3 | $1+11z+6z^2$ | y | y | 20.167 | 60.5 | 4 | 5.04 |
| harm. | 4 | $1+18z+35z^2+6z^3$ | y | y | 9.257 | 37.0 | 3 | 3.09 |
| rung2 | 4 | $1+22z+51z^2+10z^3$ | y | y | 9.490 | 38.0 | 3 | 3.16 |
| harm. | 5 | $1+37z+179z^2+133z^3+10z^4$ | y | y | 6.511 | 32.6 | 9/4 | 2.89 |
| rung2 | 5 | $1+45z+239z^2+181z^3+14z^4$ | y | y | 7.013 | 35.1 | 9/4 | 3.12 |
| harm. | 6 | $1+78z+744z^2+1286z^3+399z^4+12z^5$ | y | y | 5.518 | 33.1 | 2 | 2.76 |
| rung2 | 6 | $1+86z+920z^2+1682z^3+535z^4+16z^5$ | y | y | 5.748 | 34.5 | 2 | 2.87 |
| harm. | 7 | $1+147z+2522z^2+8948z^3+7323z^4$ $+1201z^5+18z^6$ | y | y | 4.335 | 30.3 | 16/9 | 2.44 |
| rung2 | 7 | $1+161z+3024z^2+11144z^3+9295z^4$ $+1551z^5+24z^6$ | y | y | 4.418 | 30.9 | 16/9 | 2.48 |
| harm. | 8 | $1+290z+8317z^2+51458z^3+82947z^4$ $+35270z^5+3135z^6+22z^7$ | y | y | 3.791 | 30.3 | 5/3 | 2.27 |
| rung2 | 8 | $1+298z+9277z^2+60986z^3+102275z^4$ $+44798z^5+4095z^6+30z^7$ | y | y | 3.829 | 30.6 | 5/3 | 2.30 |
| harm. | 9 | $1+559z+25519z^2+258901z^3+734059z^4$ $+631993z^5+155305z^6+8035z^7+28z^8$ | y | y | 3.293 | 29.6 | 25/16 | 2.11 |
| rung2 | 9 | $1+585z+28885z^2+305035z^3+880149z^4$ $+764143z^5+188659z^6+9789z^7+34z^8$ | y | y | 3.323 | 29.9 | 25/16 | 2.13 |
| harm. | 10 | $1+1098z+77550z^2+1213626z^3+5529648z^4$ $+8309694z^5+4180482z^6+626910z^7+19359z^8+$ $32z^9$ | y | y | 2.987 | 29.9 | 3/2 | 1.99 |
| rung2 | 10 | $1+1114z+83790z^2+1377794z^3+6457600z^4$ $+9866694z^5+5017826z^6+758710z^7+23631z^8+$ $40z^9$ | y | y | 3.004 | 30.0 | 3/2 | 2.00 |

The sweep families behave identically: every instance of $(1, 2, m)$, $m \le 14$, and $(1, 2, 3, m)$, $m \le 12$, tested by the program was real-rooted, log-concave, and Newton.

### 1.2 Strictness decay and non-correlation

Two quantitative facts were already in the program's records at $n \leq 8$ and persist through the new rows; both are stated here with the precision the data supports. First, the strictness $S$ decays monotonically in $n$ in both families, with $n \cdot S$ drifting from $\approx 37$ at $n = 3, 4$ (harmonic; the rung-2 family starts at 60.5) to $\approx 29.6$–$30.0$ at $n = 9, 10$: within the measured range the decay is well described as $\approx 30/n$. That description cannot persist indefinitely—log-concavity forces $S \geq 1$, and Newton's inequalities force $S$ above the floor $((k{+}1)(d{-}k{+}1))/(k(d{-}k)) \approx 1 + 4/d$ at the middle coefficients, so $nS \approx 30$ would cross the floor near $n \approx 26$—and much of the measured decline is the generic Newton-floor decline that any real-rooted family exhibits. The family-specific content is the excess $S$/floor, which declines from 3.06 (harmonic) and 5.04 (rung-2) at $n = 3$ to 1.99 and 2.00 at $n = 10$ (Table 1, Figure 1); it, too, declines, so there is no growing quantitative margin in the coefficient structure to spend. Second, the strictness carries no statistically distinguishable correlation with the covering margins at the achievable sample sizes: across the $n = 3$ sweep (twelve instances, $(1, 2, m)$, $m = 3..14$) the Spearman coefficient of strictness against the margin is $+0.32$ ($p = 0.32$); across the $n = 4$ sweep (eight instances, $(1, 2, 3, m)$, $m = 5..12$) it is $-0.31$ ($p = 0.45$). The sign flips between adjacent rungs and neither coefficient is significant; these arrays are too small to establish correlation or its absence. The harmonic family has margin identically 0 at every $n$ while its strictness varies by a factor of four over the table. Both coefficients were recomputed in the provenance run from the persisted instance data and reproduce exactly.

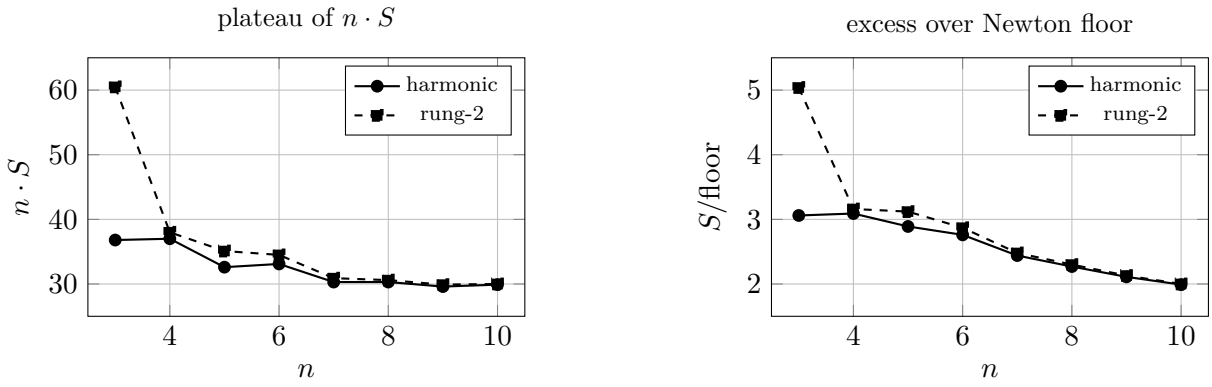



Figure 1: The strictness records of Table 1. Left: the plateau of $n \cdot S$ (the harmonic family starts at 36.8, rung-2 at 60.5; both drift to $\approx 30$). Right: the excess $S$/floor over the Newton floor at the minimizing coefficient—the family-specific content of the decline—which itself decreases from 3.06/5.04 at $n = 3$ to 1.99/2.00 at $n = 10$.

The two facts close the bridge in the coefficient direction, and the falsity of the covering-radius statement (§3.5) closes it in the metric direction: the statement that strict log-concavity would have needed to feed is false at the extremal instances, and the coefficient structure cannot see the metric hole depth that falsifies it. There is no mechanism, and the arithmetic is clean enough that the absence of the mechanism is itself a datum.

### 1.3 The residual question

What survives is a question of independent arithmetic interest, and we state it decoupled from the Lonely Runner problem entirely. The Ehrhart data of the LR-zonotopes is governed by the gcd structure (2.2) of the projected basis; the $h^*$-polynomial of the table is the ordinary Ehrhart $h^*$-polynomial of the lattice zonotope $Z(V)$ (its coefficients carry the gcd structure of the lattice geometry, but the polynomial is the ordinary one—the gcd formula (2.2) is the standard zonotope Ehrhart polynomial). In every instance tested (two core families through $n = 10$, sweeps through

$m \le 14$ at $n = 3$ and $m \le 12$ at $n = 4$), the $h^*$-polynomial is real-rooted—as it must be, by the theorem below.

**Corollary S. 1.1.** *The $h^*$-polynomial of the LR-zonotope $Z(V)$ of any speed vector $v = (v_1, \ldots, v_n)$ with distinct entries and $\gcd(v) = 1$—the Ehrhart $h^*$-polynomial of the projected-basis zonotope $\pi([0,1]^n) \subset \mathcal{X}_v$ with respect to the lattice $\Lambda = \pi(\mathbb{Z}^n) \cong \mathbb{Z}^{n-1}$—is real-rooted.*

*Proof.* $Z(V) = \pi([0,1]^n)$ is a lattice zonotope: a projection of the unit cube with vertices $\pi(\{0,1\}^n) \subset \Lambda$, i.e. a zonotope with integer generators $\pi(e_i)$ with respect to $\Lambda \cong \mathbb{Z}^{n-1}$. By Beck–Jochemko–McCullough [14], the $h^*$-polynomial of every lattice zonotope has only real roots. Hence $h^*(Z(V))$ is real-rooted. □

The real-rootedness is thus a special case of [14], and the data of Table 1 are a confirmation of that theorem on the LR-zonotopes rather than a new observation. What the table records *beyond* [14] is quantitative: the strictness $S$ (the log-concavity margin $h_k^2/(h_{k-1}h_{k+1})$), its decay, and its non-correlation with the covering margins—all properties of the coefficient sequence, not of real-rootedness alone. The data are recorded here and not pursued further: the question belongs to the Ehrhart theory of zonotopes, not to the Lonely Runner problem, and is not a bridge to it.

## 2 Computational provenance

The reproducibility standard of this paper is one sentence: *every number cited in the text traces to a script that was actually run and whose output was persisted.* The standard is applied uniformly to every cited number, inherited or new. This section is the map.

### 2.1 The ledger

The provenance run (`scripts/paper_provenance.py`; output `out_paper_provenance.json` and the human-readable `out_paper_provenance.txt`) reads the two persisted run records of the zonotope program and re-asserts, by independent exact computation, every value the paper cites from them: all witness depths (evaluating the runner formulation (2.1) *and* the independent linear-form/lattice-search implementation, asserting exact agreement of the two), all recorded $\rho$ values at their recorded argmax points, all $\lambda$ and $\delta$ values, both Spearman coefficients, both correlation instance arrays (strictness recomputed from the Ehrhart coefficients and margins recomputed from exact $\lambda$), and the strictness entries of the $n \le 8$ table. The complete ledger contains 225 items. The number of failures is 0.

The two-way witness verification deserves emphasis because it is the strongest form of check the program has: every witness value in Tables 2 and the fail-witness lists of §5 was recomputed by two implementations with different mathematics (a one-dimensional minimization over the trajectory parameter, and a pruned lattice search in the quotient norm's linear forms), agreeing exactly. A one-line reproduction of the headline witness: evaluate $\min_\sigma \max(\|\sigma\|, \|c_2 - 2\sigma\|, \|c_3 - 3\sigma\|, \|c_4 - 4\sigma\|, \|c_5 - 5\sigma\|)$ at $c = (\frac{46}{47}, \frac{92}{235}, \frac{231}{235}, \frac{35}{47})$; the minimum is attained at $\sigma = \frac{4}{235}$ and the value is $\frac{80}{235} = \frac{16}{47}$.

### 2.2 The runs behind this paper

The scripts written and run for this paper and its revisions are listed below with their persisted outputs; the inherited records are copied into the paper's artifact bundle.

1. `hstar_n9_n10.py` → `out_hstar_n9_n10.json`: the $h^*$ extension of Appendix 1 ($n = 3..10$, both families; Ehrhart asserted three ways; roots at 45 decimal places on exact integer coefficients; strictness, Newton, log-concavity exact; $\lambda$ asserted against the classical values).

2. `certify_rung2_n5.py` → `out_certify_rung2.json`: the branch-and-bound certificate of §5.4, including the cross-validation of the two $D$ implementations (forty random rational points per instance, exact agreement), the controls ($n = 3$ and $n = 4$ core instances, all certified), the negative control (harmonic $n = 5$, correctly refusing), and the ladder targets (§5).

3. `paper_provenance.py` → `out_paper_provenance.{json,txt}`: the 225-item ledger above.

4. `certify_followup.py` → `out_certify_followup.json`: the survivor enclosure analysis of the open $m = 10$ run and the recovered improved harmonic witness (607/1792 at $c = (\frac{11}{256}, \frac{163}{256}, \frac{15}{256}, \frac{79}{256})$).

5. `rho_closeout.py` → `out_rho_closeout.json`: the exact-argmax determination of the sharpest witnesses (grid and congruence-family localization of the deepest holes, exact arrangement analysis of the argmax regions, the binding-structure extraction of §5.2 with every binding value, translate, and $\chi$-residual re-computed by both $D$ implementations, and the branch-and-bound runs at the exact levels).

6. `closeout_certify.py` and `lean_closeout.py` → `out_closeout_*.json` and `out_lean_*.json`: the two-sided close-out attempts at the exact levels: the branch-and-bound survivor collection at the exact level (leaves certify $D \le r$ outside the survivor boxes), the center-plus-slack Lipschitz net over the survivors, and the exact arrangement analysis of the surviving regions (§5.4, Table 2).

7. `m9_witness_hunt.py` and `m9_ridge_scan.py` → `out_m9_witness.json`: the strict $m = 9$ witness of Table 2 (multi-start exact hill-climb, then the parametrized binding-family scan that produced the 1793/5000 point and its chain binding structure).

8. `local_refine.py`: the reusable float-grid localizer used to seed the exact arrangement analyses.

9. `certify_m10_v2.py` → `out_certify_m10_v2.json`: the $m = 10$ close-out of §5: the sound multi-slice search screen (4.2 million random points, dyadic grids, torsion neighborhood, climbs, Nelder–Mead polishes, 114 exact re-evaluations) and the re-run branch and bound with the exact Lipschitz box test added to the cascade (190,295 box evaluations, 95,148 certified leaves, zero survivors, 1,286 s; the negative control refusing, the $n = 4$ certificates re-certifying with identical counts, 846/846 sampled prunes re-confirmed exactly).

10. `certify_harmonic_v4.py` → `out_certify_harmonic_v4.json`: the harmonic close-out of §5.4: the balance-family search (the 30,000 survivor seeds of the control run, the family witness $c(35)$, the identity $32\,r_{(3,5)} + 27\,r_{(4,5)} + 35\,r_{(3,4)} = 32$ verified exactly at every member), and the branch and bound at $r = 16/47$ with the sliding-$\sigma$ gap and triple-identity certificates plus lattice-aligned splitting at the dip lattice $q = 470$ (782,255 box evaluations, 391,128 certified leaves, zero survivors, 5,935 s; 4,841/4,841 sampled prunes re-confirmed exactly); includes the $n = 6$ $m = 9$ and $m = 12$ refutation searches with their exactly evaluated dyadic witnesses.

11. `crosscheck_cert_witnesses.py`: the cross-validation of the two closure runs' witness-level claims against the independent linear-form/lattice-search implementation of this paper's ledger: seven exact evaluations (the $m = 10$ torsion center at 7/22; the balance family at $k = 20, 27, 35$, each exactly 16/47; the $m = 9$ dyadic search point at its exact depth; the $m = 12$ and $m = 9$ clean witnesses at 17/47 and 1793/5000), zero failures, plus the binding-structure extraction at the four $n = 5$-multiple center classes (pair $(1, m)$ bisecting at each), the verification behind the archetype remark of §5.2.

## 2.3 Certificates and budgets

The branch-and-bound certificates are finite objects: a certificate for $\rho(V) \le r$ is the pruned dyadic tree, each leaf carrying an exact rational $U$-value $\le r$ computed from the kink/crossover enumeration of §5.4. Budgets and certificate sizes for the runs cited in the text are listed in Table 2; the JSON persists leaf samples, survivor aggregates (count, bounding box, maximum excess over $r$, exact depths at sampled survivor centers), and the exact $U$-values, so the recorded leaves and survivor aggregates can be re-audited without re-running the search, and re-running the search reproduces every certificate deterministically (a full leaf-by-leaf re-audit of the largest runs requires the re-run). The two closure runs persist, in addition, their per-certificate fire counts (including the gap and triple-identity counts of the harmonic cascade), the sampled-prune exact re-confirmations, and the negative-control refusals; and their witness-level claims were re-asserted by an independent cross-validation run (`crosscheck_cert_witnesses.py`, seven exact evaluations, zero failures) before incorporation into this paper.

Table 2: Branch-and-bound runs cited in the text. "leaves" = certified boxes (each with exact $U \le r$, or an exact certificate of the added Lipschitz/gap/triple-identity type); "surv." = unresolved boxes at cap or at budget; wall-clock seconds in parentheses. "Direct closure" = running the tree to an empty heap at exactly $r$ with the plain cascade; it is unavailable at non-dyadic argmaxes (§5.4), which the plain-cascade rows record. "Close-out" = the two-sided assembly of leaves, the center-plus-slack Lipschitz net, and the exact region analyses (`lean_closeout.py`); "budget-open" records that the region analyses did not terminate within the stated budget. The two "certified" rows at $r = 16/47$ and at $m = 10$'s $r = 7/22$ are the closures: the enhanced cascade (gap + triple-identity, lattice-aligned splitting) and the Lipschitz box test respectively.

| instance | $r$ | verdict | leaves | surv. |
|---|---|---|---|---|
| $(1,2,3)$ | 1/4 | certified | 14 (0.0) | 0 |
| $(1,2,6)$ | 3/14 | certified | 20 (0.0) | 0 |
| $(1,2,3,4)$ | 3/10 | certified | 438 (2) | 0 |
| $(1,2,3,8)$ | 5/18 | certified | 156 (1) | 0 |
| $(1,2,3,12)$ | 7/26 | certified | 168 (1) | 0 |
| $(1,2,3,16)$ | 9/34 | certified | 222 (2) | 0 |
| $(1,2,3,4,5)$ | 1/3 | *refused* (control) | 17,520 (421) | 45,355 |
| $(1,2,3,4,5)$ | 16/47 | refused (direct closure; non-dyadic argmax) | 49,422 (3002) | 57,219 |
| $(1,2,3,4,5)$ | 16/47 | close-out (budget-open; 6,835 net-certified, 8 regions) | 42,914 (1400) | 52,581 |
| $(1,2,3,4,5)$ | 16/47 | *certified* (enhanced cascade: gap + triple-identity, lattice-aligned splitting) | 391,128 (5935) | 0 |
| $(1,2,3,4,5,12)$ | 17/47 | refused (direct closure) | 33,926 (3004) | 15,709 |
| $(1,2,3,4,5,12)$ | 17/47 | close-out (budget-open; 1,270 certified, 14 regions) | 30,340 (1400) | 9,749 |
| $(1,2,3,4,5,6)$ | 29/80 | refused (direct closure) | 52,772 (3002) | 42,822 |
| $(1,2,3,4,10)$ | 7/22 | first attempt open (residual volume $< 6 \times 10^{-7}$) | 61,038 (1501) | 21,153 |
| $(1,2,3,4,10)$ | 7/22 | *certified* (Lipschitz box test; sound search screen) | 95,148 (1286) | 0 |
| $(1,2,3,4,20)$ | 13/42 | certified | 37,018 (981) | 0 |
| $(1,2,3,4,25)$ | 4/13 | certified | 10,414 (231) | 0 |
| $(1,2,3,4,30)$ | 19/62 | certified | 10,646 (274) | 0 |

## 2.4 Reproduction

All scripts run under CPython 3.12 with `fractions`, `numpy`, `scipy`, and `mpmath`, and take minutes to hours on one core; the certificates of Table 2 take seconds to minutes (the longest plain-closure certificate, $m = 20$, took sixteen minutes; the enhanced-cascade closures took twenty-one minutes at $m = 10$ and ninety-nine minutes at the harmonic instance), and the refused control runs were

run to their stated budgets. One-line reproduction from the artifact bundle's root (each writes its `out_*.json` next to itself, deterministically):

- `python3 hstar_n9_n10.py`
- `python3 certify_rung2_n5.py`
- `python3 paper_provenance.py`
- `python3 certify_followup.py`
- `python3 rho_closeout.py` (phases: `grid47`, `tiny_exact`, `structure`, `certify`)
- `python3 lean_closeout.py 1,2,3,4,5 16/47` (and the $m = 12$, $m = 9$, harmonic-$n = 6$ analogues)
- `python3 certify_m10_v2.py` and `python3 certify_harmonic_v4.py` (the two closure runs)
- `python3 crosscheck_cert_witnesses.py` (the seven-witness cross-validation)

The artifact bundle accompanying the paper contains the scripts, their outputs, the inherited zonotope records (`out_lrc_zono_final.json`, `out_lrc_zono_rescan.json`, and the library `lrc_zono_lib.py` they import), and the ledger. Nothing in the paper relies on an unlogged computation, a hard-coded constant, or a claim of the form "verified elsewhere".

# 3 A self-contained verification core

This section verifies the sharpest negative claims self-containedly, without the companion monograph. It collects the elementary identity behind the four forms, the runner formulation with its certificate, the explicit statement of the sum-only equality, and one-line hand checks of the headline witnesses.

## 3.1 The identity and the runner formulation

**Lemma S. 3.1** (depth identity)**.** *For every $t \in \mathbb{T}$, $\max_{1\le j\le n}\|\frac{1}{2} - v_j t\| = \frac{1}{2} - \min_{1\le j\le n}\|v_j t\|$. Consequently $\delta(V) = \frac{1}{2} - \lambda(V)$ with $\delta(V) = \mathrm{dist}_\infty(\mathbb{R}v,\ \mathbb{Z}^n + \frac{1}{2}\mathbf{1})$, and the $\delta$-form of §2 is exactly* $\mathrm{LRC}_n$.

*Proof.* Each map $x \mapsto \|x\|$ satisfies $\|\frac{1}{2} - x\| = \frac{1}{2} - \|x\|$: reflecting about $\frac{1}{2}$ exchanges the two half-intervals and fixes the midpoint. Taking $x = v_j t$ and maximizing over $j$ gives the displayed identity (a maximum of quantities each equal to $\frac{1}{2}$ minus the corresponding term is $\frac{1}{2}$ minus the minimum). The distance form is the same statement along the trajectory line $tv$, $t \in \mathbb{R}$. □

The computations of this paper use the *runner formulation* (2.1), restated here:

$$D(c) \;=\; \min_{\sigma\in[0,1)}\ \max\Big(\|\sigma\|,\ \max_{2\le j\le n}\|c_j - v_j\sigma\|\Big).$$

Two facts make it auditable. First, it is the quotient-norm depth $\mathrm{dist}_{\mathcal{X}_v}(c, \Lambda)$ written in the $c$-coordinates of §2: the linear forms $\phi^{1j}$, $\phi^{ij}$ listed there generate the norm, the parameter $\sigma$ is the free coordinate $x_1$ of a lift, and minimizing over $\sigma$ minimizes over the lift. Second, it is *Lipschitz-auditable*: each term $\|c_j - v_j\sigma\|$ is $v_j$-Lipschitz in $\sigma$ and the whole objective is piecewise linear with breakpoints at the half-integers $c_j - \frac{1}{2} - v_j\sigma \in \mathbb{Z}$, so the minimum is attained at a rational breakpoint or crossing and is exactly computable by hand or by exact rational arithmetic. The independent linear-form/lattice-search implementation of the program agrees with it exactly on every instance used in this paper (forty random rational points per instance).

### 3.2 The sum-only equality, explicitly

**Theorem S. 3.2** (sum-only equality; explicit form)**.** *Let $V = (v_1, \dots, v_n)$ have distinct positive entries with $\gcd(V) = 1$ and $0 < M(V) < \frac{1}{2}$. For each unordered pair $\{u, w\} \subset [n]$ write $N_{uw} = v_u + v_w$ and*

$$\tau(u, w) \;=\; \max_{0 \le k < N_{uw}} \; \min_{p \in [n] \setminus \{w\}} \| \frac{v_p k}{N_{uw}} \|.$$

*Then $M(V) = \max_{\{u,w\}} \tau(u, w)$, and consequently $\mathrm{LRC}_n$ holds for all $V$ if and only if for every $V$ there is a pair $\{u, w\}$ with $\tau(u, w) \ge 1/(n+1)$.*

The two ingredients (proved in the companion monograph, Theorem 1.1 there; the finitization is what every instance of §3.2 entered through) are: (i) at a global optimum $t^*$ with $0 < M(V) < \frac{1}{2}$ the binding runners can be chosen two-sided—one rising, one falling—which forces $(v_u + v_w)t^* \in \mathbb{Z}$ for that pair, so $t^*$ lies on the *sum* lattice of denominator $v_u + v_w$; and (ii) on that lattice the two binding runners are exact reflections ($\|v_u k/N\| = \|v_w k/N\|$ for all $k$, since $v_u \equiv -v_w \pmod N$), so the $n$-runner minimum equals an $(n-1)$-runner minimum. The companion's Lemma 2.1 also records the boundary cases $M(V) = \frac{1}{2}$ (all speeds odd) and $M(V) = 0$ (impossible).

### 3.3 Hand checks of the headline witnesses

All the strict witnesses of Table 2 are one-minute hand computations with (2.1); the two closed values need the branch-and-bound half of their certificates as well (§5.4), but the witness halves are hand checks too.

*The harmonic argmax value* $16/47$. Take $v = (1, 2, 3, 4, 5)$ and $c = (\frac{122}{1175}, \frac{688}{1175}, \frac{284}{1175}, \frac{80}{1175})$. At $\sigma = \frac{96}{1175}$ (working in 1175-ths, with $c = (\frac{122}{1175}, \frac{688}{1175}, \frac{284}{1175}, \frac{80}{1175})$): $\|\sigma\| = \frac{96}{1175}$; $\|c_2 - 2\sigma\| = \|\frac{122-192}{1175}\| = \frac{70}{1175}$; $\|c_3 - 3\sigma\| = \|\frac{688-288}{1175}\| = \frac{400}{1175}$; $\|c_4 - 4\sigma\| = \|\frac{284-384}{1175}\| = \frac{100}{1175}$; $\|c_5 - 5\sigma\| = \|\frac{80-480}{1175}\| = \frac{400}{1175}$. The maximum is $\frac{400}{1175} = \frac{16}{47}$, attained twice (runners 3 and 5, slopes $+3$ and $-5$ opposite), and it is minimized: any local shift of $\sigma$ increases one of the two. Hence $D(c) = 16/47 > \frac{1}{3} = \mathrm{thr}_5$. The pair-form side of the same point is the quadruple binding of §5.2: $\phi^{45}(c - z) = -16/47$ at $z = (0, 0, 0, -1)$, $\phi^{34}(c - z) = -16/47$ at $z = (0, 1, 0, 0)$, $\phi^{35}(c - z) = +16/47$ at $z = 0$, and $\phi^{25}(c - z) = +16/47$ at $z = (0, 1, 1, 1)$, each a two-term integer evaluation.

*The rung-2 argmax value* $17/47$. Take $v = (1, 2, 3, 4, 5, 12)$ and $c = (\frac{4714}{5875}, \frac{1}{8}, \frac{5}{8}, \frac{6}{25}, \frac{2034}{5875})$. At $\sigma = \frac{482}{5875}$ the six terms are $\frac{482}{5875}$, $\frac{2125}{5875}$, $\frac{5693}{47000}$, $\frac{13951}{47000}$, $\frac{1000}{5875}$, $\frac{2125}{5875}$: the maximum $\frac{2125}{5875} = \frac{17}{47}$ is attained twice (runners 2 and 12, slopes $+2$ and $-12$ opposite) and is minimized, so $D(c) = 17/47 > \frac{5}{14} = \mathrm{thr}_6$. The binding triple $\{2, 5, 12\}$ and its translates are recorded in §5.2.

*The strict witnesses.* For $v = (1, 2, 3, 4, 5, 6)$ take $c = (0, \frac{5}{16}, \frac{15}{16}, \frac{3}{4}, \frac{1}{2})$ and $\sigma = \frac{23}{160}$: the six terms are $\frac{23}{160}$, $\frac{23}{80}$, $\frac{19}{160}$, $\frac{29}{80}$, $\frac{1}{32}$, $\frac{29}{80}$, so $D(c) = 29/80 > \frac{5}{14}$ (runners 4 and 6 tie at the maximum with opposite slopes). For $v = (1, 2, 3, 4, 5, 7)$ take $c = (\frac{1}{16}, \frac{9}{16}, 0, \frac{3}{8}, 0)$ and $\sigma = \frac{1}{11}$: the six terms are $\frac{1}{11}$, $\frac{21}{176}$, $\frac{51}{176}$, $\frac{4}{11}$, $\frac{7}{88}$, $\frac{4}{11}$, so $D(c) = 4/11 > \frac{5}{14}$. For $v = (1, 2, 3, 4, 5, 9)$ take $c = (\frac{1}{32}, \frac{15}{32}, \frac{1383}{10000}, \frac{32921}{40000}, \frac{95473}{200000})$ and $\sigma = \frac{18577}{200000}$: the six terms are $\frac{18577}{200000}$, $\frac{3863}{25000}$, $\frac{38019}{200000}$, $\frac{5831}{25000}$, $\frac{1793}{5000}$, $\frac{1793}{5000}$, so $D(c) = 1793/5000 > \frac{5}{14}$ (runners 5 and 9 tie at the maximum).

*The* $7/22$ *center class (the* $m = 10$ *ladder closure's witness).* Take $v = (1, 2, 3, 4, 10)$ and $c = (\frac{1}{2}, 0, \frac{1}{2}, \frac{1}{2})$. At $\sigma = \frac{7}{22}$ the five terms of (2.1) are $\frac{7}{22}$, $\frac{3}{22}$, $\frac{1}{22}$, $\frac{5}{22}$, $\frac{7}{22}$: explicitly $\|\sigma\| = \frac{7}{22}$; $\|\frac{1}{2} - 2\sigma\| = \frac{3}{22}$; $\|-3\sigma\| = \frac{1}{22}$; $\|\frac{1}{2} - 4\sigma\| = \frac{5}{22}$; and $\frac{1}{2} - 10\sigma = -\frac{59}{22} \equiv \frac{7}{22} \pmod 1$, so the fifth term is $\frac{7}{22}$. The maximum is $\frac{7}{22}$, and it is minimized: the binding terms are the $\sigma$-term (slope $+1$) and the fifth term (slope $-10$), opposite signs, so any local shift of $\sigma$ increases one of them. Hence $D(c) = 7/22$, and the branch-and-bound certificate of §5.4 ($\rho \le 7/22$, the tree emptying) upgrades this to $\rho(1, 2, 3, 4, 10) = \frac{7}{22}$ exactly; since $\delta = 1/2 - \lambda = 1/2 - 2/11 = 7/22$, the deepest hole at $m = 10$ is the center class itself. The pair-form side is the bisection of §5.2: the pair $(1, 10)$ binds at two translates with opposite signs, $2 \cdot 11 \cdot \rho = 7 \in \mathbb{Z}$.

*The balance-family witness at the harmonic argmax.* Take $c = c(35) = (\frac{46}{47}, \frac{92}{235}, \frac{231}{235}, \frac{35}{47})$. At $\sigma = \frac{4}{235}$ the five terms of (2.1) are $\frac{4}{235}$, $\frac{13}{235}$, $\frac{80}{235}$, $\frac{20}{235}$, $\frac{80}{235}$: explicitly $\|\sigma\| = \frac{4}{235}$; $\|\frac{46}{47} - 2\sigma\| =$

$\|\frac{222}{235}\| = \frac{13}{235}$; $\|\frac{92}{235} - 3\sigma\| = \frac{80}{235}$; $\|\frac{231}{235} - 4\sigma\| = \|\frac{215}{235}\| = \frac{20}{235}$; and $\|\frac{35}{47} - 5\sigma\| = \|\frac{155}{235}\| = \frac{80}{235}$. The maximum is $\frac{80}{235} = \frac{16}{47}$, attained with the third and fifth terms binding (slopes $-3$ and $+5$, opposite), and the tie over all three dips of the triple $\{3, 4, 5\}$—forced by the identity along the family—gives the same value at every member; the two-sided certificate of §5.4 then makes 16/47 the exact covering radius.

Every value in Table 2 was additionally re-asserted by two independent implementations in the provenance run (Appendix 2), and the closure runs' witnesses by the cross-validation run described there.